

SYSMON INTEGRATION AND DETECTION DEVELOPMENT IN AWS SOC HOME LAB

INTRODUCTION

This is the continuation of the previous lab which ends on splunk, Kali machine and window server with an Active directory now i moved more forward in implementation of sysmon, perform detection, alert creation and dashboard creation on my splunk.

Now let dive in sysmon log and all details as follows, Sysmon is a Windows service and device driver that, once enabled, monitors and logs system activity to the Windows event log. Its key power lies in the detail it provides. It monitors and logs system activity to the Windows Event Log (specifically Microsoft-Windows-Sysmon/Operational). Key capabilities include:

- Detailed process creation with full command lines and File hashes.
- Network connections with source/destination details.
- File creation time changes (to detect time stamping).
- Image loading, driver loads, registry modifications, DNS queries, and more.

OBJECTIVE.

The objective of this activity was to integrate Sysmon with the existing AWS SOC Lab and improve endpoint visibility by collecting detailed Windows events in Splunk for threat detection purposes.

ENVIRONMENT

- AWS EC2 Splunk Server
- Windows Server with Active Directory
- Splunk Universal Forwarder
- Sysmon
- Kali Linux Management Machine

SYSMON INSTALLATION.

Sysmon was installed on the Windows Server to provide enhanced visibility into system activities such as process creation, network connections, file creation, and DNS queries.

The steps to install firstly i downloaded sysmonconfig-export.xml from GitHub.

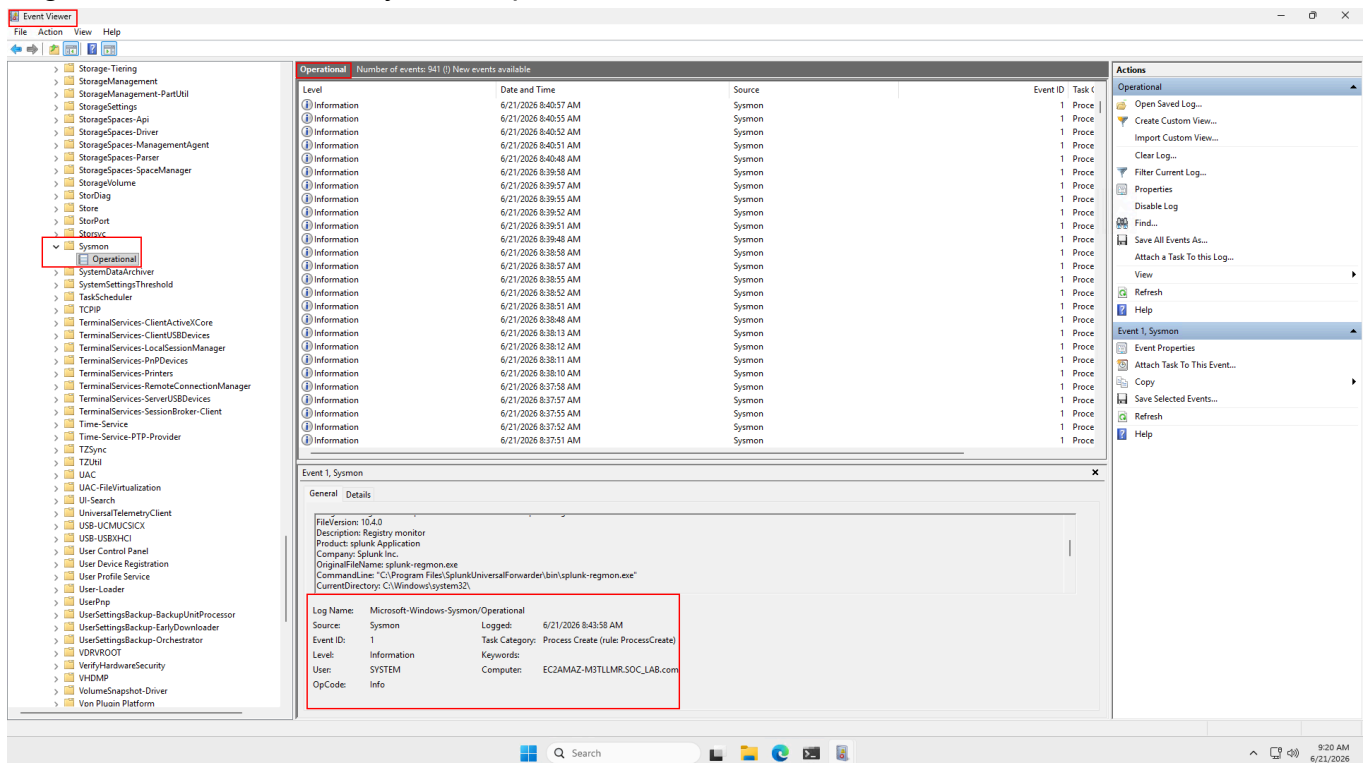
Now i install it with administrator rights

```
sysmon.exe -accepteula -i sysmonconfig-export.xml
```

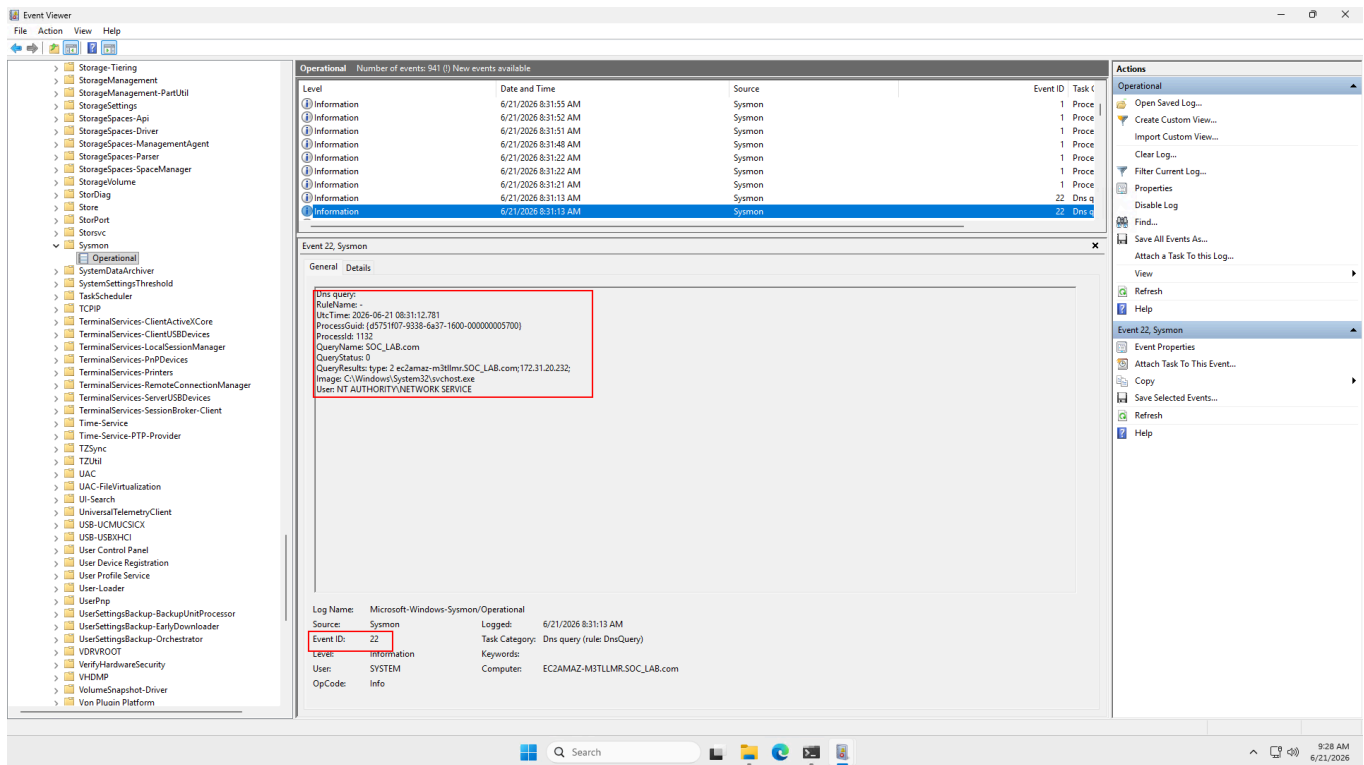
Then i update the existing configuration using.

```
sysmon.exe -c sysmonconfig-export.xml
```

So inorder to verify that sysmon installed and it work well i open the the event viewer and find the sysmon log through this path Applications and Services Logs\Microsoft\Windows\Sysmon\Operational



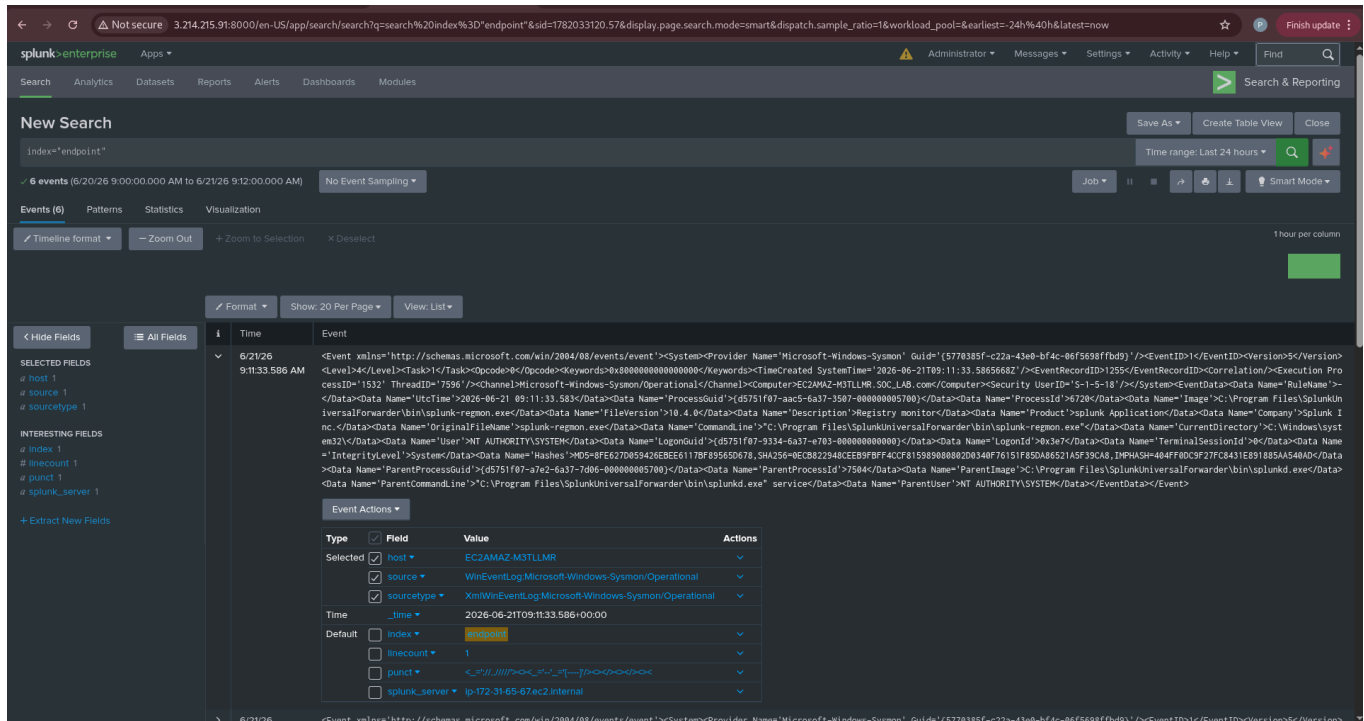
so from the image you can see those are sysmon logs.



and in this above image we see event ID=22 which shows the DNS query so this is one of the sysmon logs.

LOG FORWARDING.

The Splunk Universal Forwarder was configured to collect Sysmon logs and forward them to the Splunk server over port 9997. Hence the sysmon log accessed on splunk



Also i perform index configuration in order this sysmon logs can be accessed on splunk using the index a created for my own.

```
Splunk Universal Forwarder: Running (pid 7964)  
PS C:\Program Files\SplunkUniversalForwarder\bin> Get-Content "C:\Program Files\SplunkUniversalForwarder\etc\system\local\inputs.conf"  
[WinEventLog://Microsoft-Windows-Sysmon/Operational]  
disabled = 0  
index = endpoint  
sourcetype = XmlWinEventLog:Microsoft-Windows-Sysmon/Operational  
renderXml = true
```

Event Analysis – Sysmon Event ID 1 (Process Creation)

Sysmon Event ID 1 records every process creation event occurring on a Windows endpoint. During this analysis, the event logs were examined to understand how processes are initiated, including the relationship between parent and child processes.

The following key fields were analyzed:

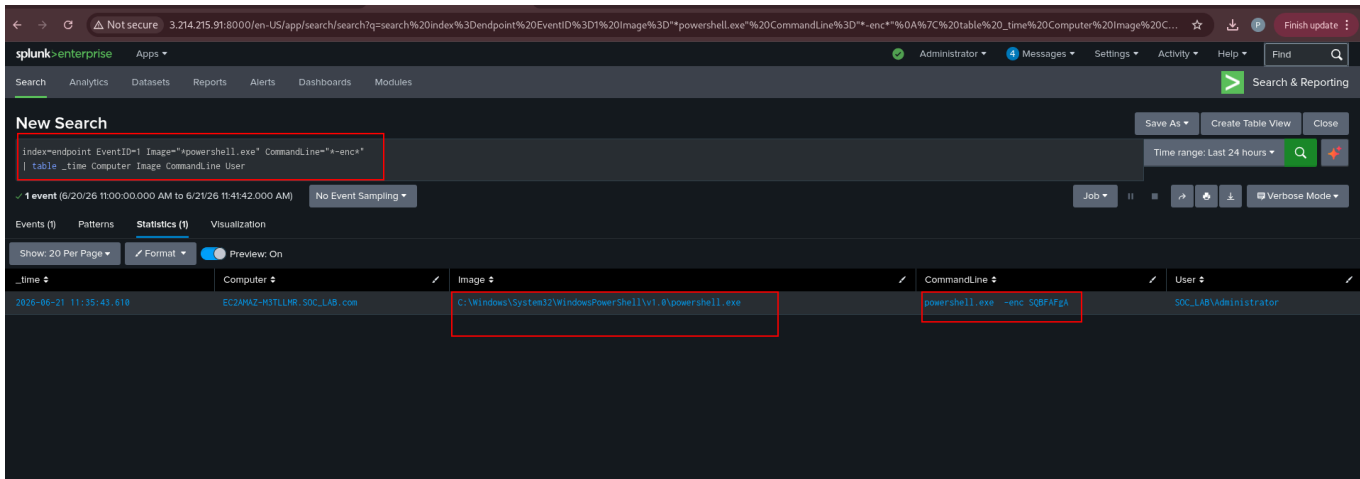
- **Process Name:** Identifies the executable that was launched (e.g., `powershell.exe`, `cmd.exe`, or other binaries).
- **Command Line:** Shows the exact command executed, which is critical for identifying malicious or suspicious activity such as encoded PowerShell commands or file downloads.
- **Parent Process:** Helps trace how the process was initiated (e.g., whether PowerShell was launched from `winword.exe`, `explorer.exe`, or another suspicious parent).
- **User Account:** Indicates which user context executed the process, helping to detect privilege abuse or unauthorized execution.
- **Process ID (PID):** Used to correlate related events such as network connections (Event ID 3) or file creation events.

During investigation, **PowerShell execution activity was observed**, which is significant because PowerShell is commonly used in both legitimate administration and attacker-driven activities such as:

- Fileless malware execution
- Script-based attacks
- Downloading payloads from remote servers
- Obfuscation using encoded commands (e.g., `-enc` or `IEX` commands)

The presence of PowerShell in Sysmon Event ID 1 required further correlation with other Sysmon logs to determine whether the activity was legitimate or potentially malicious. So the image below shows that i was able to detect a malicious activity performed using a powershell

which was a obfuscation using encoded command which was -enc.



Detection Findings

Analysis of the Sysmon logs revealed **PowerShell execution events that may indicate suspicious behavior** depending on the context of the command line.

Key observations included:

- Execution of PowerShell from a standard user or non-administrative context
- Presence of command-line arguments that may indicate automation or script execution
- Potential use of encoded or obfuscated commands (commonly used in attack scenarios)
- Parent-child process relationships that may not align with normal system behavior

Sysmon Event ID 1 proved useful in identifying these anomalies because it provides visibility into **how processes are started and what actions triggered them**.

To strengthen the detection, these events should ideally be correlated with:

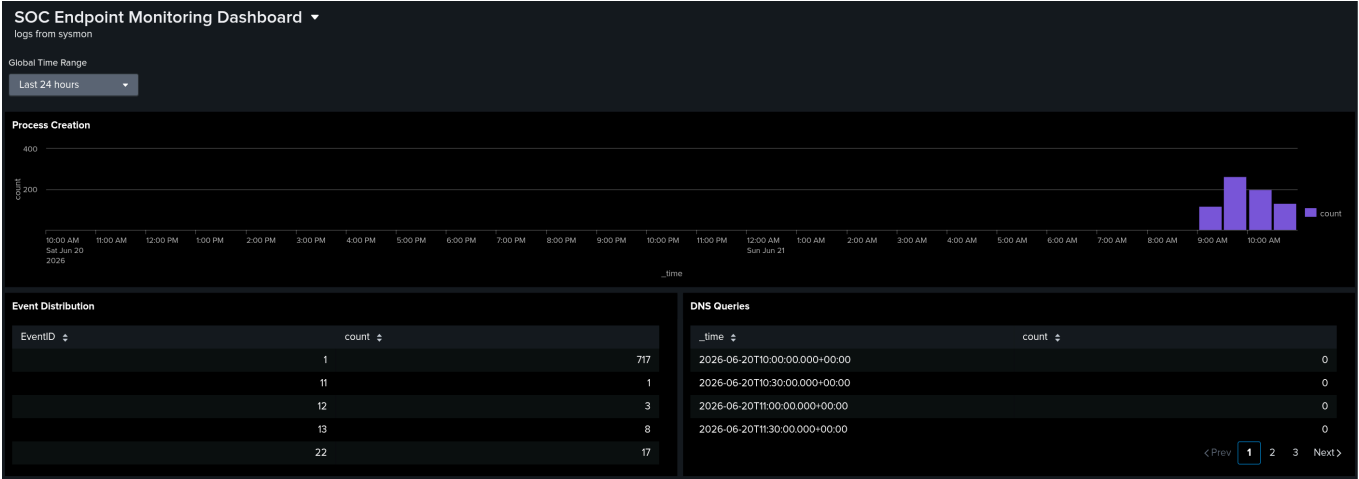
- **Event ID 3 (Network Connections):** to check if PowerShell initiated external communication
- **Event ID 11 (File Creation):** to detect dropped payloads or scripts
- **Event ID 10 (Process Access):** to identify process injection behavior

This correlation helps SOC analysts distinguish between **legitimate administrative activity and potential malicious execution chains**.

DASHBOARD AND ALERTING.

A SOC dashboard was developed in Splunk to provide a centralized view of key Sysmon telemetry, focusing on event distribution, DNS query activity, and process creation events. The event distribution panel gives an overview of all ingested Sysmon logs, helping analysts understand overall endpoint activity and identify spikes or unusual increases in logging volume. The DNS query section provides visibility into domain resolution activity from endpoints, which

is useful for detecting suspicious domains, command-and-control (C2) communication, or abnormal external lookups. The process creation dashboard (Sysmon Event ID 1) highlights executed processes, command-line arguments, and parent-child relationships to detect potentially malicious behavior such as PowerShell abuse or unauthorized execution chains. Together, these visualizations improve situational awareness and enable faster identification of anomalous activity patterns within the environment.



A Splunk alert was created to automatically detect suspicious PowerShell activity involving encoded commands. This demonstrated how SOC analysts can implement proactive monitoring and automated threat detection mechanisms.

search%20index%3Dendpoint%20EventID%3D1%20Image%3D"*powershell.exe"%20CommandLine%3D"*-enc*"&display.page.sea

Administrato

Save As Alert

Settings

Title

powershell malware

Description

Optional

Permissions

Private

Shared in App

Alert type

Scheduled

Real-time

Run every week

On

Monday

at

6:00

Expires

24

day(s)

Trigger Conditions

Trigger alert when

Number of Results

is greater than

0

Trigger

Once

For each result

Throttle

Trigger Actions

+ Add Actions

When triggered

Send email

To

changeme@example.com

Remove

Cancel

Save

3.214.215.91:8000/en-US/app/search/alert?s=%2FservicesNS%2Fadmin%2Fsearch%2Fsaved%2Fsearches%2Fpowershell%2520malware

splunkenterprise

Apps

Administrator

Messages

Settings

Activity

Help

Find

Search

Analytics

Datasets

Reports

Alerts

Dashboards

Modules

Search & Reporting

powershell malware

Edit

Enabled: Yes, Disable

App: search

Permissions: Private, Owned by admin, Edit

Modified: Jun 21, 2026 11:55:48 AM

Alert Type: Scheduled, Weekly, Monday at 6:00, Edit

Trigger Condition: .. Number of Results is > 0, Edit

Actions: 1 Action, Edit

Send email

SKILLS ACHIEVED.

Sysmon Configuration

Gained practical experience in configuring Sysmon to enhance endpoint visibility by defining specific event logging rules. This included selecting key event IDs such as process creation, network connections, and file creation events. The configuration helped ensure that only relevant security telemetry was collected, improving log quality and reducing noise during analysis.

Splunk Log Analysis

Developed the ability to ingest, search, and analyze Sysmon logs within Splunk. This included writing queries to filter events, identify patterns, and correlate different log sources. Learned how to use Splunk dashboards and searches to extract meaningful insights from raw log data for security monitoring and investigation purposes.

Process Creation Monitoring

Gained hands-on experience analyzing Sysmon Event ID 1 to monitor process execution on endpoints. This involved reviewing process names, command-line arguments, parent-child relationships, and user context to detect suspicious or abnormal behavior such as unauthorized PowerShell execution or unusual process chains.

Detection Development

Learned how to create basic detection logic based on observed patterns in logs. This includes identifying indicators of suspicious behavior, such as encoded PowerShell commands or unusual DNS queries, and translating them into detection rules or alerts within Splunk for early threat identification.

SOC Investigation Fundamentals

Developed foundational skills in SOC investigation workflows, including triaging alerts, correlating multiple log sources, and understanding attack behavior patterns. Gained insight into how analysts move from raw logs to identifying potential threats, validating incidents, and supporting incident response decisions.

FUTURE IMPROVEMENT.

- Implement advanced correlation rules.
- Integrate MITRE ATTACK mapping for detections.
- Expand Sysmon monitoring to include additional event IDs.
- Simulate more complex attack scenarios.
- Develop custom Splunk dashboards for threat hunting.

CONCLUSION

This project successfully implemented an endpoint monitoring solution using Sysmon integrated with Splunk SIEM for real-time security visibility and analysis. Sysmon was configured to capture critical endpoint activities including process creation, DNS queries, and registry modifications. The logs were successfully forwarded via Splunk Universal Forwarder and ingested into Splunk under a custom index.

Through analysis of collected telemetry, various security events were observed including normal system processes and simulated suspicious activities such as encoded PowerShell execution. This demonstrated the effectiveness of the system in capturing both benign and potentially malicious behaviors.

The integration of Sysmon and Splunk provided full visibility into endpoint activities, enabling effective detection, investigation, and analysis of system behavior from a Security Operations Center (SOC) perspective.